Advanced Computer Networks – PCS223

Assignment 8 – BGP Routing

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Objective:

To configure and analyze Border Gateway Protocol (BGP) routing between different Autonomous Systems.

Theory:

BGP is a path vector routing protocol used for communication between different Autonomous Systems (AS). It uses TCP port 179 and selects the best path based on attributes like AS_PATH and NEXT_HOP.

Topology Description:

Three routers are connected in sequence, each belonging to a different AS. Each router is connected to a PC forming separate LANs.

Router Configuration:

```
R1 (AS 100):
router bgp 100
neighbor 10.0.0.2 remote-as 200
network 192.168.1.0 mask 255.255.255.0

R2 (AS 200):
router bgp 200
neighbor 10.0.0.1 remote-as 100
neighbor 10.0.0.6 remote-as 300
network 10.0.0.0 mask 255.255.255.252

R3 (AS 300):
router bgp 300
neighbor 10.0.0.5 remote-as 200
network 192.168.3.0 mask 255.255.255.0
```

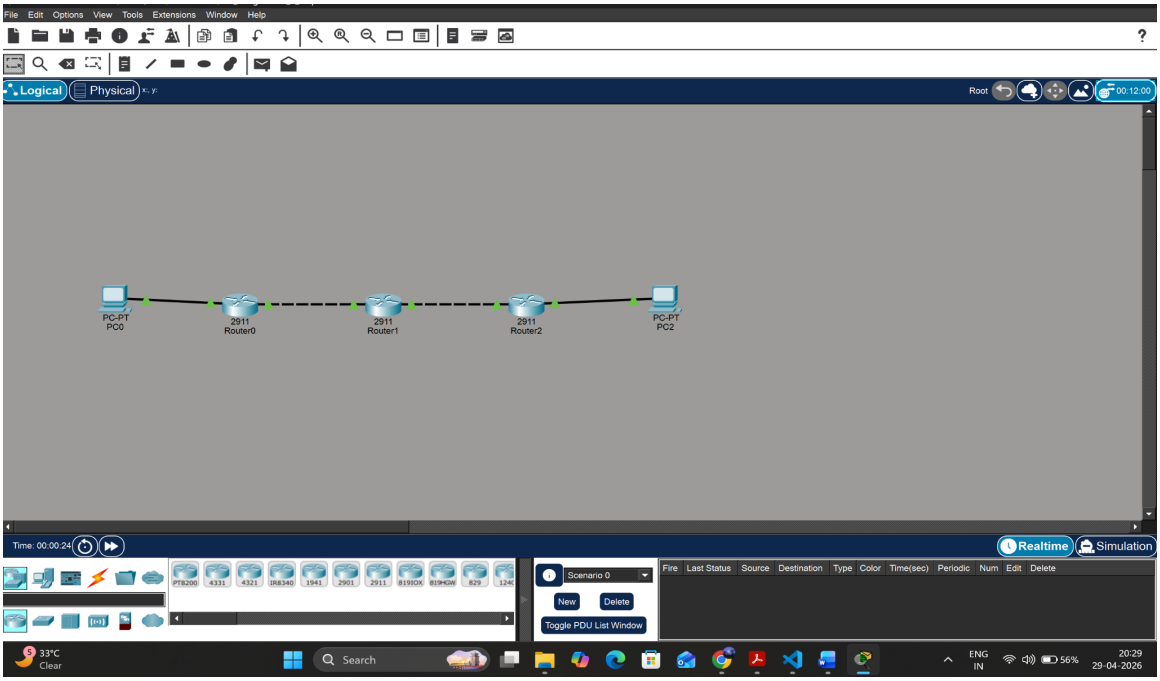
Verification:

- show ip bgp
- show ip route

• show ip bgp summary

• Ping between PCs successful

Output Screenshot:



Conclusion:

BGP routing was successfully configured between different Autonomous Systems. Routers exchanged routing information and connectivity between networks was verified.